

Presentation

“Local RAG Chatbot for customer service”

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Introduction & Context

- **Problem:** Customers frequently ask repetitive questions (vehicle features, service schedules, warranties, ordering, etc.) that are already documented internally.
- **Goal:** build a local RAG-based chatbot for automotive customer support, grounded in an internal knowledge base.
- **Idea:** Implementing a retrieval-augmented chatbot?
 - Save agent time
 - Speed up answers
 - Improve customer satisfaction by providing immediate, accurate responses.
- RAG enables LLMs to retrieve relevant, authoritative information from existing documents, keeping answers up-to-date and trustworthy.

Part 1: Setup & Knowledge Base

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Category	Advantages (Pros)	Disadvantages (Cons)
Cost & Setup	Free & Open Source: no monthly “pod” costs (e.g., Pinecone Pricing).	Single-Node Architecture: Difficult to scale horizontally; if you outgrow one server, you must migrate.
Speed	Great for Prototyping: fastest way to build a working demo.	Concurrency Bottleneck: High traffic causes “database is locked” errors due to the SQLite backend.
Performance	Local Storage: All data stays on your machine for maximum privacy.	Memory Dependency: Because of the HNSW algorithm, the entire index must fit in your System RAM.

Scalable Alternatives

- **Weaviate:** Best for enterprise-grade scalability due to **Horizontal Sharding**, **Hybrid Search** and Smart **Filtering** support.
- **FAISS:** Best for maximum raw search speed due to **GPU** support and advanced search algorithms, e.g. Inverted File (**IVF**).

Part 1: Setup & Knowledge Base

LLM and Embedding models



Llama3.2 3B params

- Runs on a laptop CPU. At 3B params the model fits in ~2 GB of RAM.
- Instruction-following is enough for summarizing texts rather than to generating ones.
- Temperature = 0. Deterministic output means the same question always gives the same answer — essential for a support chatbot.
- Customer queries stay fully local, no external API.

nomic-embed-text

- Nomic is lightweight (274MB).
- Top MTEB benchmark scores.
- Long 8192-token context window.
- Entirely local.
- Same Ollama runtime. One tool manages both models.

Document Ingestion

Optimizing Chunk Parameters: 25% Semantic Overlap

Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

 duration of standard warranty?



 The duration of the standard warranty is 2 years.

Sources:

- Warranty Coverage (filename: warranty_coverage.txt)

 Sources

 when customer line is available?



 According to our online support channels, the Live Chat on our website is available during business hours.

Sources:

- Online Support: Live Chat on our website during business hours
- Customer Support Channels

 Sources

CHUNK_SIZE = 300

CHUNK_OVERLAP = 75

Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

 duration of standard warranty?



 The duration of the Standard Warranty is 2 years with unlimited mileage.

Sources:

- Warranty Coverage (Electric Vehicle Battery Warranty section)

 Sources

 when customer line is available?



 The Customer Hotline is available from Monday to Friday, 8:00 - 20:00, and on Saturday, 9:00 - 17:00.

Sources:

- Customer Support Channels (Hotline Availability)

 Sources

CHUNK_SIZE = 400

CHUNK_OVERLAP = 100

Document Ingestion

Optimizing Chunk Parameters: 15% Semantic Overlap

Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

 duration of standard warranty?



 The duration of the standard warranty is 2 years.

Sources:

- Warranty Coverage (filename: warranty_coverage.txt)

 Sources



 when customer line is available?



 According to our online support channels, the Live Chat on our website is available during business hours.

Sources:

- Online Support: Live Chat on our website during business hours
- Customer Support Channels

 Sources



CHUNK_SIZE = 300

CHUNK_OVERLAP = 45

Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

 duration of standard warranty?



 The duration of the Standard Warranty is 2 years with unlimited mileage.

Sources:

- Warranty Coverage (Electric Vehicle Battery Warranty section)

 Sources



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 The Customer Hotline is available from Monday to Friday, 8:00 - 20:00, and on Saturday, 9:00 - 17:00.

Sources:

- Customer Support Channels (Hotline Availability)

 Sources



CHUNK_SIZE = 400

CHUNK_OVERLAP = 60

Category	Advantages (Pros)	Disadvantages (Cons)
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Speed	Fast to Prototype	Concurrency Bottleneck: High traffic causes “database is locked” errors due to the SQLite backend
Performance	Local Storage	Memory Dependency: The entire index must fit in your system RAM (HNSW)

Scalable Alternatives

- **FAISS:** Best for maximum raw search speed due to **GPU** support and advanced search algorithms, e.g., Inverted File (**IVF**).
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Part 2: RAG Pipeline

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Overview

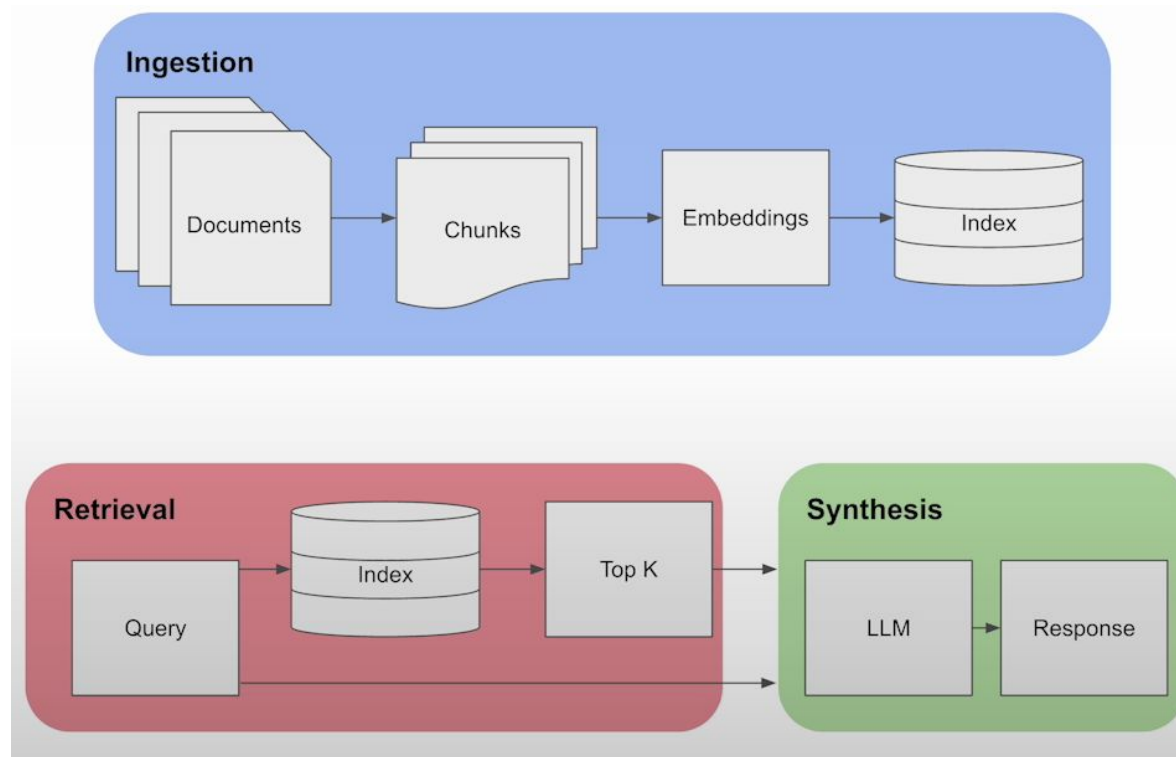


Figure 1: Basic RAG Pipeline

Part 2: RAG Pipeline

Multi-Stage Retrieval & Refinement



1. **Query Expansion (Multi-Query):** LLM generates three variations of the input to improve semantic coverage.
2. **Ensemble Retrieval:** Parallel search across Vector (Semantic) and BM25 (Keyword) stores, merged via Reciprocal Rank Fusion (RRF).
3. **Contextual Compression:** Post-processing to extract only the relevant snippets from the top chunks, reducing noise and token usage.
4. **Re-Ranking (Cross-Encoder):** High-precision scoring of query-document pairs to move the most relevant chunks to the top.

Part 2: RAG Pipeline

Enhanced Retrieval Pipeline



Technique	What it fixes
BM25 (keyword)	Embeddings miss exact keyword matches (model names)
EnsembleRetriever	Fuses keyword and semantic search chunks via Reciprocal Rank Fusion
MultiQueryRetriever	Rephrase the question 3 times
Contextual Compression + Reranker	Filter irrelevant sentences from retrieved chunks before sending to LLM

Table 1: Overview of the Retrieval Stage and implemented techniques being used.

Part 2: RAG Pipeline

Response Generation

```
SYSTEM_PROMPT = """\nYou are a helpful and professional customer support assistant for a large automotive company.\n\nAnswer the customer's question using ONLY the information in the Context section below.\n\nRules:\n- If the context contains the answer: give a clear, helpful response.\n- If the context does NOT contain the answer: say exactly –\n  "I'm sorry, I don't have that information in my knowledge base.\n  Please contact our support team at customer.service@bmw.com or call 089 125016000."\n- Never invent facts or use knowledge outside the provided context.\n- End every answer with a "Sources:" section listing the document filename(s) you used.\n\nContext:\n{context}\n"""
```

System Prompt:

- **Safe Fallback:** Ensures the LLM does not hallucinate answers.
- **Authentic Contact Info:** Provides the official email and customer service phone number of **BMW**.
- **Source Attachment:** Attaches the document(s) used by the model.
- **PromptTemplate Usage:** To generate responses.

Part 3: Demo in Action

Settings

Retrieved chunks (Top-K)

1 3 10

How it works

1. Your question is embedded
2. Relevant document chunks are retrieved
3. An LLM generates an answer based on the context

Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

 duration of standard warranty?

 The duration of the Standard Warranty is 2 years with unlimited mileage.
Sources: Warranty & Guarantees [Warranty Coverage]

Sources

 When the customer line is available?

 The Customer Hotline is available Monday to Friday, 8:00 - 20:00. On Saturdays, it's available from 9:00 - 17:00.
Sources:

- Customer Support Channels [Customer Service]

Sources

Part 4: Tools & Evaluation

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How well does the chatbot perform?

Metric	Score	Status	Interpretation
Avg Faithfulness	1.00 / 1.0	✓ Excellent	The LLM is strictly using the provided context and not hallucinating.
Avg Relevance	1.00 / 1.0	✓ Excellent	The generated answers directly address the user's specific questions.
Avg Context Recall	0.60 / 1.0	⚠ Warning	The retriever is missing about 40% of the relevant information needed for a full answer.

Works well:

- Perfect faithfulness and relevance across all 6 questions.
- The system prompt, temperature 0, and reranker together prevent hallucinations and keep answers on-topic.
- The LLM never invents facts or goes off-topic.

What to improve:

- Context recall drops on multi-step answers (warranty, ordering).
- The reference uses different phrasing than the retrieved chunks.
- Next step: expand the eval set to 49 questions and tune chunk size.

Part 4: Tools & Evaluation



LangChain & LangSmith

✓	Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation Queue	Tokens
✓	RunnableSequence	what is the duration o...	The duration of th...		3/3/2026, 9:5...	🕒 30.83s			593
✓	ContextualCompressionRetriever	what is the duration o...	[{"metadata":{"ca...		3/3/2026, 9:5...	🕒 11.24s			171

LangSmith is LangChain's observability platform. It's very useful for debugging why a particular answer was wrong.

What to improve: Latency.

Part5: Conclusion & Outlook

Conclusion:

- We introduce RAGChatbot, a Q&A system built for fast prototyping using ChromaDB and Llama 3.2 3B.
- To improve retrieval, we use multiple retrievers with re-ranking.
- Optimal settings: 400 token chunk size with 60-token overlap (15%).
- Integrates LangSmith with LangChain for pipeline orchestration.

Future Work:

- **Improve scalability** by migrating to different vector stores, e.g., **FAISS**, **WEVIATE**.
- **Semantic Routing**: The system "routes" the question based on its meaning/complexity.
- Analyze and improve the embeddings space using **PCA**, investigate how the query and similar documents are located in the high-dimensional space.
- **Optimized RAG**: Implement **LlamaIndex** to accelerate local document processing and reduce latency in response generation.
- **Recursive Querying**: Enable the bot to follow internal document references (e.g., "See Section 2.1") automatically for more complete answers.
- Implement advanced techniques of **Prompt Engineering**:
 - **Prompt Sanitization**: cleans user inputs from malicious attacks.
- **Metadata filtering**: remove irrelevant documents.
- **Improve chunking strategy**
- Create a **Docker** image and deploy it to production.

The background of the slide is a vibrant green with a series of concentric, curved lines that create a sense of motion and depth, resembling a stylized landscape or a dynamic graphic element.

Thank you for your attention!